

Zhenyao Cui

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Research Interests

Transfer learning; cross-modal alignment; neural decoding.

Education

Huazhong University of Science and Technology

Sep 2024 – Jun 2027

M.Eng. in Intelligent Science and Technology (expected)

- GPA **3.89/4.00**; selected for direct graduate admission through the national recommendation scheme.
- Advisor: Prof. **Dongrui Wu**, IEEE Fellow and Editor-in-Chief of *IEEE Transactions on Fuzzy Systems*.

Huazhong University of Science and Technology

Sep 2020 – Jun 2024

B.Eng. in Automation, Elite Engineer Program

- GPA **3.91/4.00**; Outstanding Graduate of HUST.

Publications

[†] equal contribution.

1. **Zhenyao Cui**, Siyang Li, Siyuan Kan, Yudi Wang, Fei Luo, and Dongrui Wu. “A Fuzzy Set Based Classification-to-Regression Extension Framework for Transfer Learning and Its Application to Brain-Computer Interfaces.” *IEEE Transactions on Emerging Topics in Computational Intelligence*, Early Access, 2026.
doi:10.1109/TETCI.2026.3683713
2. Siyang Li[†], Jiayi Ouyang[†], **Zhenyao Cui**[†], Ziwei Wang, Tianwang Jia, Feng Wan, and Dongrui Wu. “Backpropagation-Free Test-Time Adaptation for Lightweight EEG-Based Brain-Computer Interfaces.” *IEEE Journal of Biomedical and Health Informatics*, accepted for publication, 2026.
arXiv:2601.07556
3. Dingkun Liu[†], Yuheng Chen[†], Zhu Chen[†], **Zhenyao Cui**, Yaozhi Wen, Jiayu An, Jingwei Luo, and Dongrui Wu. “EEG-FM-Compass: Progress, Benchmarking, and Future Directions for EEG Foundation Models.” *National Science Review*, nwag466, 2026.
doi:10.1093/nsr/nwag466
4. **Zhenyao Cui**, Siyuan Kan, Siyang Li, Ziwei Wang, and Dongrui Wu. “SCORE: Subject Coordinate Recovery for Label-Free Cross-Subject EEG-to-Image Retrieval.” Preprint; manuscript under review, 2026. arXiv:2608.19134.
5. **Zhenyao Cui**, Siyuan Kan, Dingkun Liu, and Dongrui Wu. “Beyond Trial Averaging: Anchoring Neural and Visual Representations for Few-Repetition Brain-to-Image Retrieval.” Preprint; manuscript under review, 2026. arXiv:2608.19128.
6. **Zhenyao Cui**, Jiahui Chai, Siyuan Kan, Yudi Wang, and Dongrui Wu. “MCL-Align: Marginal-Conditional-Label Alignment for Cross-Subject EEG Regression.” Under review, *IEEE Transactions on Biomedical Engineering*, 2026.
7. Siyuan Kan, Huanyu Wu, **Zhenyao Cui**, Fan Huang, Xiaolong Xu, and Dongrui Wu. “CMCRD: Cross-Modal Contrastive Representation Distillation for Emotion Recognition.” *IEEE Transactions on Affective Computing*, 17(2):1837–1846, 2026.
doi:10.1109/TAFFC.2026.3664341

Research Experience

Transferable EEG Decoding

Apr 2024 – Present

Laboratory of Brain-Computer Interfaces and Machine Learning, HUST

- Proposed **C2R**, a fuzzy-set framework that extends class-conditional transfer learning from classification to continuous EEG regression, and derived a target-domain risk bound based on fuzzy class-conditional probabilities. (*IEEE TETCI, Early Access, 2026*)
- Co-authored **EEG-FM-Compass**; contributed to the literature review and dataset curation, and evaluated the regression performance of multiple EEG foundation models, including **LaBraM** and **CBraMod**, on the SEED-VIG continuous PERCLOS prediction task under cross-subject LOSO and 10% within-subject calibration, comparing full fine-tuning with linear probing using RMSE and correlation. (*National Science Review, 2026*)
- Proposed **MCL-Align**, a comprehensive probability-alignment framework that jointly aligns marginal, conditional, and label distributions through Euclidean Alignment, Conditional Direction Matching, and label standardization. (*IEEE TBME, under review*)
- As equal-first author, co-developed **BFT**, which combines knowledge-guided EEG augmentation, Monte Carlo-dropout-inspired feature masking, and a source-trained learning-to-rank module to weight transformed predictions, suppress test-time uncertainty, and improve transfer without backpropagation. (*IEEE JBHI, accepted, 2026*)

Brain-to-Image Retrieval

Dec 2025 – Present

- Identified systematic cross-subject coordinate differences in EEG-to-image retrieval: subjects preserve similar concept relationships but express them along different directions. Proposed **SCORE** to select reliable EEG-image landmarks and estimate an orthogonal reorientation for label-free coordinate recovery.
- Identified a non-transitive alignment pattern in few-repetition retrieval: the low-repetition neural query and image representation each align with the high-repetition center, but not with each other. Proposed **NEAR** to use this center as an anchor, denoising queries and predicting pseudo-anchors for gallery images across EEG, MEG, and fMRI.

Selected Projects and Competitions

SSVEP-Controlled Robotic Arm with Adaptive Step Size

Sep 2025 – Dec 2025

- Built an SSVEP acquisition and decoding system for fast, precise robotic-arm control, using position-aware adaptive step sizing to balance movement speed and endpoint precision; won the **national championship** in the Brain-Controlled Robotic Arm track of the 2025 China Brain-Computer Interface Competition. The system was later deployed at Wuhan Optics Valley Biotech City; the competition was covered by CCTV and *Global Times*.

Porcine ECoG-Based Continuous Joint-Motion Decoding

2024

- Developed a porcine ECoG-based continuous joint-motion decoding algorithm with a segment-slicing augmentation method tailored to recurring joint-motion patterns during running; won **First Prize** in the minimally invasive BCI track at the 2024 World Robot Contest Championship.

Industry Experience

Alibaba Cloud Intelligence Group

Apr 2026 – Present

Research Intern

- Developed a foundation embedding model specialized for Elastic Compute Service (ECS) logs by designing a domain-specific tokenizer and contrastive pretraining objectives, then pairing it with a downstream classifier that integrates temporal attention. On production server-log data, the system improved F1 from **0.4383** to **0.5306** (+21.1% relative).
- Contributed to Alibaba Cloud's internal **AutoResearch** platform. Given a problem definition, a code branch, and evaluation metrics, the platform drives an agent to iteratively modify the solution, run evaluations, and improve task performance.

Dongfeng Motor Group — Multimodal Intelligent Cockpit Evaluation

Jun 2025 – Jun 2026

Industry-funded project — Project Lead

- Developed a multimodal intelligent-cockpit evaluation framework integrating EEG, images, audio, and text to assess users' subjective experience of in-vehicle functions.

Selected Awards and Honors

- National Champion**, China Brain-Computer Interface Competition, Brain-Controlled Robotic Arm track, 2025.
- First Prize**, Porcine ECoG-Based Continuous Joint-Motion Decoding, World Robot Contest Championship, 2024.
- First Prize**, National Undergraduate Intelligent Vehicle Competition, Multi-Vehicle Formation track, 2022.
- First-Class Graduate Academic Scholarship, HUST, 2024–25 and 2025–26; Yunnan Scholarship, 2024–25; Huawei Scholarship, 2021–22.
- Outstanding Graduate of HUST, 2024.